

Zhengkao (Daniel) Yu

Zhengkao@connect.hku.hk|+86 18057036888|Pokfulam, Hong Kong SAR

EDUCATION

The University of Hong Kong

BEng in Computer Engineering

- GPA: **3.40/4.3 (Top 15%, 2558 students)**
- *Relevant Coursework: Probabilistic Systems Analysis, Data Structures and Algorithms, Operating Systems, Embedded Systems, Digital System Design, Pattern Recognition and Machine Intelligence, etc*

Hong Kong SAR

Sept 2023–Jun 2027

Peking University

Academic Exchange Program

- *Relevant Coursework: Simulation Methods for Optimization and Learning (A)*

Beijing, China

Jun 2024–Aug 2024

PUBLICATION

- **Zhengkao Yu** (First Author). *Design and Implementation of a Wheeled Robot with Obstacle Avoidance and Navigation Capabilities for Drug Delivery*. Published in 2025 International Conference on Electronics, Electrical and Grid Technology (ICEEGT 2025), pp. 704–720.
- **Zhengkao Yu** (First Author). *Game-Diffusion-based Multi-Agent Learning of Ramp Merging Coordination for Connected and Automated Vehicles*. *IEEE Open Journal of Intelligent Transportation Systems (OJITS)*, 2026 (Accepted/In Press).
- **Zhengkao Yu** (First Author). *Design and Wizard-of-Oz Evaluation of a Tri-Modal Contactless In-Vehicle HMI for High-Noise and High-Vibration Cabins*. Accepted in *Journal of Intelligence and Knowledge Engineering (JIKE, ISSN: 2959-0620)*.

RESEARCH EXPERIENCE

Human – Computer Interaction Research Program

Independent Research

Remote

Jan 2026–Present

- Explored contactless in-vehicle human-computer interaction systems for off-road and engineering machinery under high-noise and high-vibration conditions
- Proposed a lightweight multimodal interaction paradigm integrating gaze selection, head confirmation, and noise-robust speech refinement
- Designed a three-layer perception – understanding – feedback architecture for multimodal fusion and conflict resolution
- Developing an interactive prototype for mainstream engineering vehicles, and proposing evaluation models for cognitive load and operational efficiency.

Adversarial Testing for Autonomous Driving

Research Assistant, supervised by Prof. Hao Zhang (CMU & UTA)

Remote

Jun 2025–Present

- Investigated systematic failure discovery in autonomous driving planning systems under safety-critical highway merging scenarios
- Formulated merging interactions as a non-cooperative game and developed a multi-agent reinforcement learning (MARL) framework integrated with diffusion-based generative models to synthesize adversarial traffic behaviors
- Generated 2000+ valid safety-critical scenarios in HighwayEnv (PyTorch implementation); achieved 92% effectiveness in triggering unexpected planning behaviors
- Designed SOTIF-aligned evaluation protocols; improved failure detection rate by 85% compared to random testing
- Reduced the collision risk of the tested planning module from 12.3% to 3.7% through targeted adversarial validation

Robust Autonomous Navigation for Wheel-Leg Hybrid Robot

Remote

Independent Research

Apr 2025–Jul 2025

- Designed and validated a hybrid wheel-leg robotic platform for autonomous drug delivery in constrained hospital environments
- Developed an EKF-based multi-sensor fusion framework integrating LiDAR, IMU, ultrasonic, and infrared sensors for robust localization under dynamic obstacles
- Proposed an integrated motion planning architecture combining JPS-optimized A* global planning with DWA local obstacle avoidance for real-time adaptation
- Implemented PID-based self-balancing control (tilt error $\pm 2^\circ$) and multi-mode locomotion
- Validated through real-world hospital experiments: 98.2% obstacle avoidance success rate, ≤ 5 cm localization error, ≤ 10 cm path tracking error, 0.3 s dynamic obstacle response

INDUSTRY EXPERIENCE

SevnceRobotics

Shanghai, China

Electrical Engineering Intern, Product R&D Center

Jun 2025 – Aug 2025

- Designed dual-layer CAN architecture (CAN FD/CAN C) and 24-series lithium battery BMS for humanoid robotic platforms
- Reduced communication latency by 12% and improved anti-interference robustness in industrial deployment scenarios
- Leveraged Altium Designer to complete schematic design and 4-layer PCB layout for sensor acquisition board; achieved 99.9% data transmission success rate after physical debugging; used MATLAB for system simulation and reliability testing
- Cross-collaborated with Control Algorithm, Product Testing and Procurement teams to align technical requirements, formulate test cases and output preliminary BOM; participated in HR160 electrical architecture V0.3 design reviews and proposed adopted optimization improvements

ADDITIONAL INFORMATION

Technical Skills:

- Programming: Python (PyTorch/OpenCV), C/C++ (Embedded/ROS), MATLAB/Simulink, Shell Script, Verilog/SystemVerilog, SQL
- Tools: Altium Designer, CANoe, Git/GitHub, VS Code, LaTeX
- ML: PyTorch (MARL/Diffusion), ROS 2, EKF/UKF, PID, A*/JPS/DWA, OpenCV

Interests: Robotics Architecture, Robotics System Design, Autonomous Driving (Planning & Testing), Multimodal HCI, Embedded Systems

Languages: Chinese (Native), English (Fluent, IELTS 7)